



Cambridge International AS & A Level

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MATHEMATICS**9709/13**

Paper 1 Pure Mathematics 1

October/November 2025**1 hour 50 minutes**

You must answer on the question paper.

You will need: List of formulae (MF19)

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.

INFORMATION

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [].

This document has **16** pages.



- 1 (a) Expand $\left(2 - \frac{1}{2}x\right)^6$ in ascending powers of x up to and including the term in x^3 . [3]

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- (b) Hence find the coefficient of x^3 in the expansion of $(3 - x + 2x^3)\left(2 - \frac{1}{2}x\right)^6$. [2]

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- 3 The equation of a curve is $y = f(x)$, where $f(x) = \frac{1}{2}x^{\frac{2}{3}}(x-2)^2$. The following points lie on the curve. Non-exact values of the y -coordinates are given correct to 6 decimal places.

$$A(8, 72), B(8.001, k), C(8.01, 72.300388), D(8.1, 75.038882)$$

- (a) Find the value of k . Give your answer correct to 6 decimal places. [1]

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The table below shows the gradients of the chords AB and AC , given correct to 4 decimal places.

Chord	AB	AC	AD
Gradient of chord	30.0039	30.0388	

- (b) Find the gradient of the chord AD . Give your answer correct to 4 decimal places. [1]

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- (c) State what the values in the table suggest about the value of $f'(8)$. [1]

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4 The first, second and third terms of a progression are 20, k and $k - 5$ respectively.

(a) Given that the progression is arithmetic, find the 30th term.

[2]

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(b) Given instead that the progression is geometric, find the sum to infinity.

[4]

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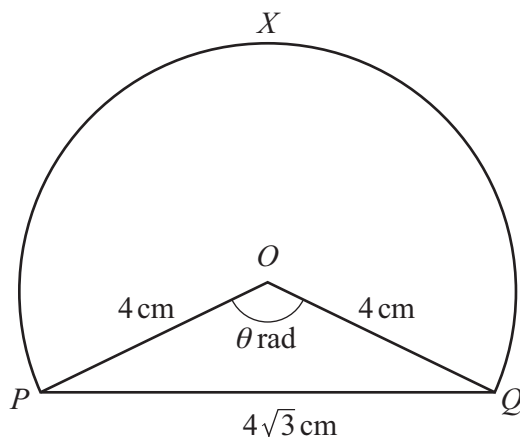
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The diagram shows part of a circle with centre O and radius 4 cm. The chord PQ is of length $4\sqrt{3}$ cm and angle $POQ = \theta$ radians. The point X lies on the circle.

- (a) Find the exact value of θ . [2]

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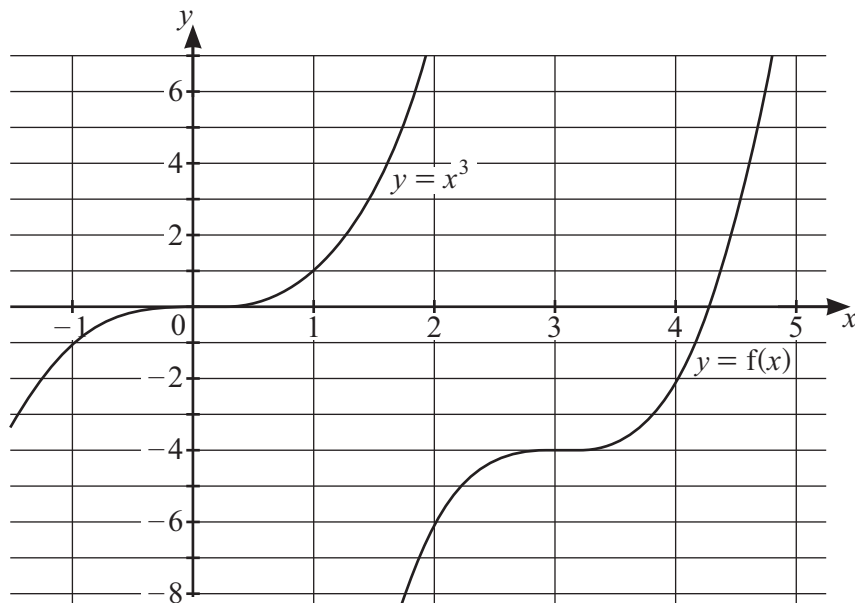
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- (b)** Find the exact area of the segment PXQ . [3]

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6



The diagram shows the graphs of $y = x^3$ and $y = f(x)$. The graph of $y = x^3$ is transformed to the graph of $y = f(x)$ by a sequence of transformations.

- (a) Describe fully a suitable sequence of transformations. Make clear the order in which the transformations are applied. [5]

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- (b) You are given that $f(x) = a(x+b)^3 + c$.

State the values of the constants a , b and c .

[3]

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8 The points $(6, 1)$ and $(-2, 7)$ lie at the opposite ends of a diameter of a circle.

(a) Find the equation of the circle.

[3]

[illegible]

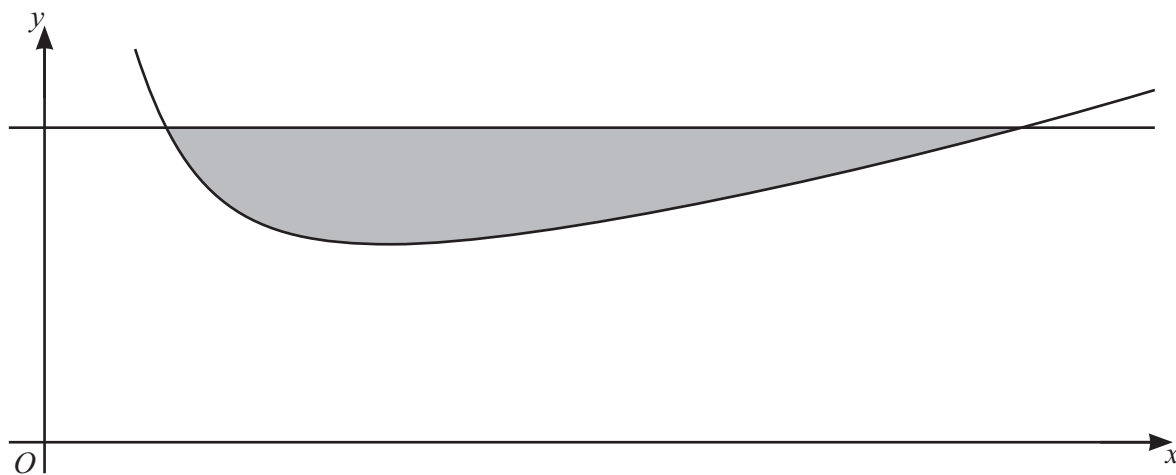
(b) There are two tangents to the circle which have gradient $-\frac{1}{2}$.

Find the exact values of the x -coordinates of the points at which these tangents touch the circle.

[5]

This image shows a full page of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page, providing a template for handwriting practice or general writing. There are no margins, text, or other markings on the page.

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The diagram shows part of the curve with equation $y = \frac{1}{2}x + \frac{4}{x}$ and the line $y = 4.5$.

Find the exact volume of the solid formed when the shaded region is rotated through 360° about the x -axis. [8]

This image shows a full page of a document template designed for handwriting practice. It features 18 evenly spaced, horizontal dashed lines across the entire width of the page. The background is plain white, providing a clear guide for letter height and placement. There are no margins, text, or other markings present.



DO NOT WRITE IN THIS MARGIN

Handwriting practice area with 20 sets of horizontal dotted lines.





10 A function f is defined by $f(x) = px^2 + 4x + q$ for $x \in \mathbb{R}$, where p and q are constants.

(a) It is given that $p = 2$ and $q = 10$.

(i) Express $f(x)$ in the form $a(x+b)^2 + c$, where a , b and c are constants.

[3]

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(ii) State the range of f .

[1]

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(b) It is given instead that $q = -5$ and the roots of $f(x) = 0$ are $5m$ and $-9m$, where m is a constant.

Find the values of p and m .

[5]

[illegible]

(b) (i) Find the x -coordinates of each of the stationary points of the curve.

[3]

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(ii) Find $\frac{d^2y}{dx^2}$ and hence determine the nature of each of the stationary points.

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If you use the following page to complete the answer to any question, the question number must be clearly shown.

[illegible]

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